Supplementary Materials

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Supplementary Materials

Reliability of Personality Measures

Overview

Reliability was evaluated for both the full baseline PID-5 questionnaire (220 items) and the brief EMA-based PID-5 assessment (15 items, 3 per domain). For the EMA measures, which involve repeated assessments nested within persons, we employed multilevel reliability estimation following Lai (2021), which distinguishes between-person reliability (consistency of person-level means) from within-person reliability (consistency of occasion-level deviations).

Baseline PID-5 Questionnaire

The full PID-5 was administered at baseline (T1) using the standard 220-item version. Reliability was computed using Cronbach's alpha and McDonald's omega for each domain and for the total score. Results are reported for both the full sample and a cleaned sample excluding participants who failed attention check items (catch trials embedded at positions 68 and 161 in the questionnaire).

Results. All domains showed acceptable to excellent internal consistency (Table S1). Psychoticism exhibited the highest reliability (α = .87, ω = .90), consistent with its larger item pool (33 items). Antagonism showed the lowest reliability (α = .67, ω = .82), though McDonald's omega—which accounts for item heterogeneity—indicated adequate composite reliability. The total PID-5 score demonstrated excellent reliability (α = .96, ω = .98). Excluding participants who failed attention checks produced slightly lower but comparable estimates, indicating that careless responding had minimal impact on scale properties in this sample.

Table S1

Internal consistency of baseline PID-5 domains.

Domain	Items	α (Full)	ω (Full)	α (Clean)	ω (Clean)
Negative Affectivity	23	0.783	0.851	0.767	0.843
Detachment	22	0.754	0.844	0.741	0.840
Antagonism	21	0.670	0.819	0.627	0.796
Disinhibition	22	0.747	0.843	0.725	0.824
Psychoticism	33	0.870	0.898	0.864	0.892
Total PID-5	220	0.965	0.979	0.964	0.977

EMA-Based Brief PID-5

The brief PID-5 administered via EMA comprised 15 items (3 per domain), selected based on factor loadings and domain representativeness from prior validation work (Bottesi et al., 2024). Because these items were assessed repeatedly across approximately 20 occasions per participant, standard single-level reliability estimates are inappropriate. Instead, we employed the multilevel composite reliability framework described by Lai (2021), which partitions variance into within-person and between-person components and computes separate reliability indices for each level.

Multilevel Reliability Framework. Following Lai (2021), we estimated three reliability indices:

- α_{2L} **(Two-level alpha)**: Overall reliability of observed scores, pooling within- and between-person variance. Relevant when scores are used without distinguishing levels.

- α_B **(Between-person alpha)**: Reliability of person-level means (aggregated across occasions). This is the relevant index when EMA scores are averaged to create a single trait estimate per person, as in our moderation analyses.

- α_W **(Within-person alpha)**: Reliability of occasion-level deviations from each person's mean. Relevant for detecting momentary fluctuations in personality states.

The same decomposition was applied using McDonald's omega (ω_{2L}, ω_B, ω_W), which relaxes the assumption of tau-equivalence.

Results. The multilevel reliability analysis revealed a clear pattern (Table S2):

1. **Between-person reliability was high** (α_B = .87, ω_B = .85). This indicates that person-level trait estimates derived from aggregating across EMA occasions are measured with good precision. This is the critical index for our moderation analyses, which use person-level latent trait scores as predictors.

2. **Within-person reliability was adequate** (α_W = .73, ω_W = .72). This suggests that the brief scale can detect meaningful occasion-to-occasion fluctuations in personality states, though with more measurement error than at the between-person level. This is expected given the brevity of the scale (15 items total).

3. **Two-level reliability was good** (α_{2L} = .83, ω_{2L} = .81), reflecting the overall quality of observed scores when levels are pooled.

Table S2

Multilevel reliability of EMA-based brief PID-5 (15 items).

Reliability Index	Estimate	95% CI Lower	95% CI Upper
α_{2L} (two-level)	0.826	0.802	0.845
α_B (between)	0.866	0.835	0.889
α_W (within)	0.727	0.687	0.758
ω_{2L} (two-level)	0.814	—	—
ω_B (between)	0.852	—	—
ω_W (within)	0.724	—	—

Interpretation for the Present Study. The high between-person reliability (α_B = .87) supports the validity of using EMA-derived personality scores as person-level moderators of vocal stress responses. Because our moderation model estimates latent trait scores from the repeated EMA observations (see Statistical Models section), measurement error is explicitly modeled rather than ignored. The multilevel measurement model in our Stan implementation can be viewed as formalizing the reliability structure documented here: the latent trait θ_{id} represents the "true" person-level standing on domain d, while the occasion-specific observations X_{nd} are treated as noisy indicators with residual variance σ_d^{ema} capturing within-person fluctuation and measurement error.

The adequate within-person reliability (α_W = .73) also suggests that the brief EMA measure could support analyses of state-level personality-voice covariation, though such analyses would require denser sampling than the present design to achieve adequate statistical power.

Descriptive Statistics

Descriptive statistics are reported for $N = 119$ female participants. Participants contributed on average 27 EMA assessments ($SD = 4.16$, range = 12–31).

Design and Coverage

Table S3 summarizes the study design, including the number of participants, voice observations, and EMA assessments. All 119 participants contributed usable voice observations in each assessment period (Baseline, Pre-exam, Post-exam), consistent with the balanced, three-phase design.

Table S3

Study design and data coverage.

N	EMA *M*	EMA *SD*	EMA Range	Voice Obs. (per period)
119	27	4.16	12–31	119

Definition and psychological significance of the three principal acoustic features

Table S4 summarizes the definition, physiological basis, and psychological significance of the three principal acoustic features discussed in this literature.

Cross-vowel convergence

Cross-vowel convergence of F0. Because the primary F0 outcome was computed as the mean of the three sustained vowels (/a/, /i/, and /u/), we conducted an additional cross-vowel convergence analysis to verify that the F0 signal was not driven by one specific vowel. The analysis included the same 119 participants used in the main F0 models, with complete F0 data for all three vowels at baseline, pre-exam, and post-exam assessments, yielding 357 voice observations.

Table S4

Principal Acoustic Features of Speech: Definitions and Psychological Significance

Feature	Definition	Physiological basis	Psychological significance
Fundamental frequency (F0)	Rate of vocal fold vibration (Hz); acoustic correlate of perceived pitch	Cricothyroid and vocalis muscle tension regulates vocal fold stiffness and vibratory rate	Most robust acoustic marker of stress: F0 increases reliably under acute stress, cognitive load, and evaluative threat via sympathetic activation
Normalized Noise Energy (NNE)	Ratio of inharmonic to harmonic spectral energy (dB); index of glottal noise	Degree of glottal closure completeness during phonation	Reflects phonatory control and vocal effort; stress may increase noise (incomplete closure under arousal) or decrease it (compensatory hyperadduction and pressed phonation)
Jitter[a]	Cycle-to-cycle variation in F0 period (%); measure of frequency perturbation	Irregularity of successive vocal fold vibratory cycles	Proposed as an index of vocal instability and reduced neuromuscular control; elevated in some emotional and clinical states

Note. F0 and NNE were retained as primary outcome variables in the present study. [a]Jitter was extracted but not included in the main analyses for the following reasons: (a) the literature on jitter and psychological stress is inconsistent, with effect sizes that are small, heterogeneous in direction, and often non-significant across studies (Giddens et al., 2013); (b) jitter is highly sensitive to recording conditions, signal-to-noise ratio, and extraction algorithm parameters, resulting in substantial measurement noise in field settings (Titze, 1994); and (c) unlike F0 and NNE, which index theoretically distinct mechanisms (autonomic arousal vs. phonatory control), jitter does not map onto a clearly separable psychophysiological pathway, limiting its interpretive value in the context of personality moderation hypotheses.

First, we examined the pairwise associations among F0 values extracted from /a/, /i/, and /u/. Across all timepoints pooled, the three vowels were strongly associated, with Spearman correlations ranging from .87 to .89. The same pattern was observed within each timepoint: correlations ranged from .84 to .89 at baseline, from .86 to .92 pre-exam, and from .88 to .90 post-exam. Thus, absolute F0 values showed high convergence across the three sustained vowels.

Second, we examined whether F0 change scores were also consistent across vowels. Stress-related changes, defined as PRE − BASELINE, were moderately to strongly associated across vowels, with Spearman correlations ranging from .64 to .83. Recovery-related changes, defined as POST − PRE, showed a similar pattern, with correlations ranging from .69 to .80. Total changes from BASELINE to POST were also convergent across vowels, with correlations ranging from .63 to .77. These results indicate that the temporal F0 response was not specific to a single vowel.

Third, we estimated the reliability of the cross-vowel F0 composite. The consistency reliability of the three-vowel mean was high for absolute F0 values, ICC(3,k) = .97, 95% CI [.96, .97]. Reliability remained high when computed separately by timepoint: ICC(3,k) = .97 at baseline, .95 pre-exam, and .98 post-exam. Reliability was also adequate for the change-score composites: ICC(3,k) = .83, 95% CI [.77, .88], for PRE − BASELINE changes; ICC(3,k) = .82, 95% CI [.76, .87], for POST − PRE changes; and ICC(3,k) = .87, 95% CI [.83, .91], for POST − BASELINE changes.

Finally, as a sensitivity check, we fitted a mixed-effects model in which F0 was predicted by the stress and recovery contrasts, vowel, and the interactions between vowel and the two time contrasts, with random intercepts for participant and observation. Adding the vowel-by-time interactions did not significantly improve model fit relative to a model without these interactions, $^{2}(4) = 7.70$, $p = .103$. The estimated PRE − BASELINE F0 increase was positive for all three vowels: /a/ = 3.77 Hz, /i/ = 2.73 Hz, and /u/ = 3.94 Hz.

Overall, these analyses support the use of the cross-vowel F0 mean as a robust aggregate. F0 values were highly concordant across /a/, /i/, and /u/, and stress-related F0 changes showed moderate-to-strong cross-vowel convergence. The primary F0 findings therefore do not appear to be driven by a single sustained vowel.

Cross-vowel convergence of NNE. Because the primary NNE outcome was computed as the mean of the three sustained vowels (/a/, /i/, and /u/), we conducted an additional cross-vowel convergence analysis to evaluate whether the NNE signal generalized across vowels or was primarily driven by a single vowel. The analysis included 141 participants with complete NNE data for all three vowels at baseline, pre-exam, and post-exam assessments, yielding 423 voice observations.

First, we examined pairwise associations among NNE values extracted from /a/, /i/, and /u/. Across all timepoints pooled, cross-vowel Spearman correlations were moderate, ranging from .40 to .67. The same pattern was observed within timepoint: correlations ranged from .30 to .67 at baseline, from .46 to .69 pre-exam, and from .41 to .63 post-exam. Thus, absolute NNE values showed moderate convergence across vowels, with the strongest associations generally observed between /i/ and /u/.

Second, we examined whether NNE change scores were also consistent across vowels. Stress-related changes, defined as PRE − BASELINE, showed moderate cross-vowel convergence, with pairwise Spearman correlations ranging from .30 to .47. Recovery-related changes, defined as POST − PRE, showed a similar pattern, with correlations ranging from .27 to .46. Total changes from BASELINE to POST were also moderately associated across vowels, with correlations ranging from .26 to .42. These results indicate that NNE change scores were not vowel-specific, although cross-vowel convergence was weaker than for F0.

Third, we estimated the reliability of the three-vowel NNE composite. The consistency reliability of the cross-vowel mean was acceptable for absolute NNE values, ICC(3,k) = .75, 95% CI [.71, .79]. Reliability was similar when estimated separately by timepoint: ICC(3,k) = .71 at baseline, .79 pre-exam, and .74 post-exam. Reliability for change-score composites was lower but still acceptable: ICC(3,k) = .65, 95% CI [.54, .74], for PRE − BASELINE changes; ICC(3,k) = .70, 95% CI [.60, .78], for POST − PRE changes; and ICC(3,k) = .60, 95% CI [.48, .71], for POST − BASELINE changes.

Finally, as a sensitivity check, we fitted a mixed-effects model in which NNE was predicted by the stress and recovery contrasts, vowel, and the interactions between vowel and the two time contrasts, with random intercepts for participant and observation. Adding the vowel-by-time interactions did not significantly improve model fit relative to a model without these interactions, $^2(4) = 4.15$, $p = .387$. The estimated PRE − BASELINE NNE change was negative for all three vowels, although its magnitude varied across vowels: /a/ = −1.05, /i/ = −0.18, and /u/ = −0.72.

Overall, these analyses support the use of the cross-vowel NNE mean as a reasonable aggregate. NNE showed moderate cross-vowel convergence, acceptable reliability of the three-vowel composite, and no evidence that the temporal NNE pattern differed significantly by vowel. At the same time, convergence was weaker than for F0, suggesting that NNE may be more vowel-sensitive and that NNE-based findings should be interpreted with somewhat greater caution.

Voice Outcomes by Assessment Period

Tables S5 and S6 report descriptive statistics for the two primary voice outcomes—fundamental frequency (F0) and normalized noise energy (NNE)—across the three assessment periods.

Across periods, mean F0 increased from Baseline (191.1 Hz) to Pre-exam (194.8 Hz) and decreased slightly at Post-exam (192.6 Hz), consistent with anticipatory stress-related

Table S5

Voice outcome descriptives by assessment period.

Period	*n*	F0 *M* (Hz)	F0 *SD*	NNE *M* (dB)	NNE *SD*
Baseline	119	191.11	21.30	-26.43	2.63
Pre-exam	119	194.81	21.36	-27.12	3.25
Post-exam	119	192.63	23.12	-26.93	2.87

elevation in pitch followed by partial recovery. NNE showed more negative values (indicating reduced glottal noise) from Baseline to Pre-exam, with partial return toward Baseline in the Post-exam period.

Table S6 summarizes within-person variability across the three assessment periods, reporting the distribution of participant-specific standard deviations.

EMA Personality Domain Descriptives

EMA-based PID-5 domain descriptives are reported separately for between-person variability (Table S7) and within-person variability (Table S8).

Between-person descriptives indicate substantial inter-individual variability in EMA trait levels, while within-person descriptives confirm meaningful intra-individual fluctuation across repeated assessments. This pattern is consistent with the multilevel measurement model used to estimate latent trait scores and propagate measurement uncertainty into moderation effects.

Table S6

Within-person variability in voice parameters across periods.

Variable	n	Mean SD	SD of SDs	Min SD	Max SD
F0 within-person SD (Hz)	119	7.44	5.47	0	28.76
NNE within-person SD (dB)	119	1.73	1.11	0	5.58

Statistical Models

Main Effects of Exam-Related Stress on Vocal Parameters

All analyses used hierarchical Bayesian models implemented in Stan via the rstan package, with four chains of 4,000 iterations each (2,000 warmup). Convergence was verified through R-hat statistics (all < 1.01) and trace plot inspection. We report posterior medians with the probability of direction (pd) as the primary index of evidence and 89% equal-tailed credible intervals (CrIs) as summaries of magnitude uncertainty; see *Inferential Framework and Reporting* below.

Research Question. Before examining personality moderation, we first establish whether exam-related stress produces reliable changes in vocal acoustics. This model addresses the foundational question: Do vocal parameters (F0, NNE) change systematically across the three assessment phases—baseline, pre-exam (anticipatory stress), and post-exam (recovery)?

Inferential Framework and Reporting. All analyses were conducted within a Bayesian estimation framework, and we report and interpret full posterior distributions rather than dichotomous accept/reject decisions about individual parameters. Our inferential aim was to characterize the *direction*, *relative strength*, and *uncertainty* of the estimated associations and to interpret the overall pattern of results across parameters. For an exploratory study with a modest sample, this is the scientifically informative output of the analysis, whereas a binary verdict on each parameter would discard most of that

Table S7

EMA PID-5 descriptives: between-person variability (person means across occasions).

Domain	n	M	SD	Min	Max
Negative Affectivity	119	4.53	1.62	0.71	8.08
Detachment	119	2.04	1.62	0.00	6.19
Antagonism	119	0.87	1.17	0.00	5.29
Disinhibition	119	2.46	1.29	0.10	5.80
Psychoticism	119	1.25	1.44	0.00	5.73

information and invite the very significance-testing logic that a Bayesian treatment is meant to supersede (Kruschke & Liddell, 2018; McElreath, 2018).

We summarize each parameter with three complementary quantities.

1. **Posterior median** — a point summary of the most probable parameter value given the model and the data.

2. **Probability of direction (pd)** — the posterior probability that the parameter takes its more probable sign, $pd = max[P(\theta > 0), P(\theta < 0)]$ (Makowski, Ben-Shachar, Chen, & Lüdecke, 2019). The *pd* is our primary index of evidence. It ranges from .50 (the sign is completely uncertain) to 1.00 (the sign is effectively certain) and has a direct verbal reading: a *pd* of .97 means that, given the model and data, there is a 97% posterior probability that the effect lies in the stated direction. This is a probability statement *about the parameter*, which is available only in the Bayesian framework; under the frequentist framework the parameter is a fixed unknown and no probability can be attached to it. We treat the *pd* as a *continuous* index of directional evidence and deliberately impose no threshold above which an effect is declared "present"; we report it for every parameter and interpret the

Table S8

EMA PID-5 descriptives: within-person variability (participant SDs across occasions).

Domain	n	M of SD	SD of SDs	Min SD	Max SD
Negative Affectivity	119	1.32	0.42	0.57	2.42
Detachment	119	1.20	0.62	0.00	3.04
Antagonism	119	0.71	0.65	0.00	2.84
Disinhibition	119	1.17	0.45	0.30	2.31
Psychoticism	119	0.93	0.68	0.00	2.93

gradient of values.

3. **89% credible interval (equal-tailed)** — the central 89% of the posterior, reported as a summary of the *magnitude uncertainty* of the estimate. This interval is **not** a significance test: whether or not it includes zero carries no special inferential status, and we do not interpret it that way. The 89% width (rather than 95%) is chosen deliberately and follows McElreath (2018): Any interval width is arbitrary, and using a value other than 95% signals that the number is a descriptive convenience rather than a decision boundary, discouraging the reflexive mapping of "the 95% interval excludes zero" onto "$p < .05$." A 50% interval, or any other width, would serve the same descriptive purpose; what matters is the location and spread of the posterior mass, not a cutoff.

Two consequences of this framework are worth making explicit. First, **we draw conclusions from the pattern of results across the full set of parameters, not from the status of any single estimate.** The claim that personality moderation of vocal stress responses is domain- and phase-specific rests on the *joint* configuration of directional probabilities across the ten F0 and ten NNE moderation parameters, not on any one interaction crossing a threshold. Second, **direction is better identified than**

magnitude. As shown in the prior-sensitivity analysis (Supplement, *Prior-Sensitivity Analysis*), the sign and directional probability of the moderation effects are robust across prior specifications, whereas their point magnitudes are only weakly constrained by the present sample. We therefore foreground direction and directional evidence (*pd*) and treat all reported magnitudes — and the widths of their credible intervals — as estimates carrying substantial uncertainty (Gelman & Carlin, 2014).

We did not use Bayes factors or region-of-practical-equivalence (ROPE) decision rules. Bayes factors reintroduce a thresholded, model-selection logic and are sensitive to the prior on the parameter under test — undesirable here, since the prior-sensitivity analysis shows the moderation magnitudes to be prior-dependent. A ROPE would require specifying a region of practical equivalence in the original vocal units (Hz, dB), for which there is no established basis in this novel measurement context. The probability of direction provides a more transparent and assumption-light summary of the evidence for our exploratory, directional questions.

Model Specification. The model is a standard Bayesian hierarchical (multilevel) linear regression with repeated measures nested within participants. Stress effects are parameterized using two orthogonal contrasts:

- **Stress contrast (c_1)**: Compares pre-exam to baseline, capturing anticipatory stress-induced change.
- **Recovery contrast (c_2)**: Compares post-exam to pre-exam, capturing post-stressor trajectory.

Let y_n denote the vocal outcome for observation n, with participant index $s[n]$. The model specifies:

$$y_n \sim \text{Normal}(\mu_n, \sigma_y)$$

where the linear predictor includes both fixed and random effects:

$$\mu_n = \alpha + u_{0,s} + (\beta_1 + u_{1,s}) \cdot c_{1n} + (\beta_2 + u_{2,s}) \cdot c_{2n}$$

The parameters are:

- α: Grand intercept (population-average baseline vocal level).
- β_1: Population-average stress effect (pre-exam vs. baseline).
- β_2: Population-average recovery effect (post-exam vs. pre-exam).
- $u_{0,s}$: Participant-specific random intercept.
- $u_{1,s}$: Participant-specific random slope for stress.
- $u_{2,s}$: Participant-specific random slope for recovery.
- σ_y: Residual standard deviation.

The random effects capture individual differences in baseline vocal characteristics (u_0), stress reactivity (u_1), and recovery patterns (u_2). A non-centered parameterization is used for computational efficiency:

$$u_{k,s} = \tau_k \cdot z_{k,s}, \quad z_{k,s} \sim \text{Normal}(0, 1)$$

where τ_k are the random effect standard deviations.

Prior Specification. Priors were chosen to be weakly informative, incorporating domain knowledge about plausible parameter ranges while allowing the data to dominate inference (Table S9).

Interpretation. The key parameters of interest are β_1 (stress effect) and β_2 (recovery effect):

- A positive β_1 indicates that F0 increases from baseline to pre-exam, consistent with

Table S9

Prior specifications for the main effects model.

Parameter	Prior	Rationale
α	Normal(220, 30)	Centered on typical female F0 (Hz)
β_1, β_2	Normal(0, 10)	Weakly informative; allows effects up to $\pm$20 Hz
τ_k	Exponential(0.5)	Weakly informative for random effect SDs
σ_y	Exponential(0.1)	Allows residual SD in plausible range

heightened autonomic arousal elevating vocal pitch.

- A negative β_2 would indicate that F0 decreases from pre-exam to post-exam, suggesting recovery toward baseline levels.

The random effect standard deviations (τ_1, τ_2, τ_3) quantify the degree of individual differences in baseline levels, stress reactivity, and recovery patterns. Large values of τ_2 or τ_3 would indicate substantial heterogeneity in how participants respond to stress—heterogeneity that might be explained by personality traits, motivating the moderation analyses presented in the next section.

The `generated quantities` block produces posterior predictive samples for model checking and pointwise log-likelihoods for model comparison via LOO-CV.

Supplementary Sensitivity Analysis for Main Effects: Robust Student-t Models

To assess whether the main findings were sensitive to the Gaussian residual assumption and to the influence of potentially outlying observations, we fitted robust versions of the main-effects models for both acoustic outcomes: mean fundamental frequency, F0, and normalized noise energy, NNE. The robust models retained the same fixed-effect structure as the primary models, with two planned contrasts: stress, comparing PRE with BASELINE, and recovery, comparing POST with PRE. The hierarchical structure was also retained, but the subject-level random intercepts and random slopes were modeled as correlated random effects. The residual likelihood was changed from Gaussian to Student-t with fixed degrees of freedom, $\nu = 4$, providing heavier tails and therefore reduced sensitivity to influential observations.

The robust models were fitted as sensitivity analyses rather than as entirely separate substantive models. Thus, the key question was whether the direction, magnitude, and uncertainty of the stress and recovery effects remained consistent with the original Gaussian models, and whether the robust specification improved predictive diagnostics.

```
// ===========================================================================
// f0_main_effects_student_t_corr.stan
// Hierarchical Bayesian model for F0 mean
// Main effects of stress (PRE vs BASELINE) and recovery (POST vs PRE)
// Robust Student-t likelihood + correlated random effects
// ===========================================================================

data {
  int<lower=1> N_subj;
  int<lower=1> N_obs;
  array[N_obs] int<lower=1, upper=N_subj> subj_id;
  vector[N_obs] y;
  vector[N_obs] c1;
  vector[N_obs] c2;
}
transformed data {
  real<lower=2> nu = 4;
}
```

```
parameters {
  real alpha;
  real b1;
  real b2;
  // Correlated random effects, non-centered parameterization
  matrix[3, N_subj] z_u;
  vector<lower=0>[3] tau;
  cholesky_factor_corr[3] L_Omega;
  real<lower=0> sigma_y;
}
transformed parameters {
  // u[s, 1] = random intercept for subject s
  // u[s, 2] = random slope for c1 for subject s
  // u[s, 3] = random slope for c2 for subject s
  matrix[N_subj, 3] u;
  u = (diag_pre_multiply(tau, L_Omega) * z_u)';
}
model {
  alpha ~ normal(220, 30);  // typical F0 scale, weakly informative
  b1 ~ normal(0, 10);       // stress effect
  b2 ~ normal(0, 10);       // recovery effect
  tau ~ exponential(0.5);
  L_Omega ~ lkj_corr_cholesky(2);
  to_vector(z_u) ~ std_normal();
  sigma_y ~ exponential(0.1);
  for (n in 1:N_obs) {
    int s = subj_id[n];
```

```
    real mu = alpha + u[s, 1]
            + (b1 + u[s, 2]) * c1[n]
            + (b2 + u[s, 3]) * c2[n];
    y[n] ~ student_t(nu, mu, sigma_y);
  }
}
generated quantities {
  corr_matrix[3] Omega;
  vector[N_obs] y_rep;
  vector[N_obs] log_lik;
  vector[N_subj] log_lik_subj;
  Omega = multiply_lower_tri_self_transpose(L_Omega);
  log_lik_subj = rep_vector(0, N_subj);
  for (n in 1:N_obs) {
    int s = subj_id[n];
    real mu = alpha + u[s, 1]
            + (b1 + u[s, 2]) * c1[n]
            + (b2 + u[s, 3]) * c2[n];

    y_rep[n] = student_t_rng(nu, mu, sigma_y);
    log_lik[n] = student_t_lpdf(y[n] | nu, mu, sigma_y);
    log_lik_subj[s] += student_t_lpdf(y[n] | nu, mu, sigma_y);
  }
}
```

```
// ===========================================================================
// nne_main_effects_student_t_corr.stan
```

```
// Hierarchical Bayesian model for Normalized Noise Energy (NNE)
// Main effects of stress (PRE vs BASELINE) and recovery (POST vs PRE)
// Robust Student-t likelihood + correlated random effects
// ==========================================================================

data {
  int<lower=1> N_subj;
  int<lower=1> N_obs;
  array[N_obs] int<lower=1, upper=N_subj> subj_id;
  vector[N_obs] y;
  vector[N_obs] c1;
  vector[N_obs] c2;
}
transformed data {
  real<lower=2> nu = 4;
}
parameters {
  real alpha;
  real b1;
  real b2;
  // Correlated random effects, non-centered parameterization
  matrix[3, N_subj] z_u;
  vector<lower=0>[3] tau;
  cholesky_factor_corr[3] L_Omega;
  real<lower=0> sigma_y;
}
transformed parameters {
```

```
  // u[s, 1] = random intercept for subject s
  // u[s, 2] = random slope for c1 for subject s
  // u[s, 3] = random slope for c2 for subject s
  matrix[N_subj, 3] u;
  u = (diag_pre_multiply(tau, L_Omega) * z_u)';
}
model {
  // Priors for fixed effects
  alpha ~ normal(-28, 5);  // typical NNE values around -20 to -30 dB
  b1 ~ normal(0, 5);       // stress effect
  b2 ~ normal(0, 5);       // recovery effect
  tau ~ exponential(0.2);
  L_Omega ~ lkj_corr_cholesky(2);
  to_vector(z_u) ~ std_normal();
  sigma_y ~ exponential(0.2);
  for (n in 1:N_obs) {
    int s = subj_id[n];
    real mu = alpha + u[s, 1]
            + (b1 + u[s, 2]) * c1[n]
            + (b2 + u[s, 3]) * c2[n];
    y[n] ~ student_t(nu, mu, sigma_y);
  }
}
generated quantities {
  corr_matrix[3] Omega;
  vector[N_obs] y_rep;
  vector[N_obs] log_lik;
```

```
  vector[N_subj] log_lik_subj;
  Omega = multiply_lower_tri_self_transpose(L_Omega);
  log_lik_subj = rep_vector(0, N_subj);
  for (n in 1:N_obs) {
    int s = subj_id[n];
    real mu = alpha + u[s, 1]
            + (b1 + u[s, 2]) * c1[n]
            + (b2 + u[s, 3]) * c2[n];
    y_rep[n] = student_t_rng(nu, mu, sigma_y);
    log_lik[n] = student_t_lpdf(y[n] | nu, mu, sigma_y);
    log_lik_subj[s] += student_t_lpdf(y[n] | nu, mu, sigma_y);
  }
}
```

We report posterior medians, MADs, and central 89% credible intervals, computed as the 5.5th and 94.5th posterior percentiles.

F0 model. For F0, the original Gaussian model estimated a positive stress effect, with a median effect of 3.27 Hz, MAD = 1.25, and an 89% credible interval from 1.25 to 5.27 Hz. The posterior probability that the stress effect was positive was 0.995. In the robust Student-t model, the estimated stress effect was smaller, with a median of 2.30 Hz, MAD = 1.14, and an 89% credible interval from 0.47 to 4.16 Hz. The posterior probability that the stress effect was positive remained high, pd = 0.979.

Thus, the robust model attenuated the estimated F0 stress effect by approximately 0.97 Hz, but the effect remained positive, with a high probability of direction (pd = 0.979).

For the recovery contrast, the original Gaussian model showed essentially no clear effect, median = 0.14 Hz, MAD = 1.24, 89% CrI [-1.89, 2.11], with $P(b2 > 0) = 0.542$. The robust model estimated a slightly negative recovery effect, median = -0.39 Hz, MAD = 1.14, 89% CrI [-2.21, 1.41], with $P(b2 > 0) = 0.368$. This result remains compatible with no clear recovery effect.

Outcome	Model	Parameter	Median	MAD	89% CI	Directional probability
F0	Gaussian	Stress, b1	3.27	1.25	[1.25, 5.27]	$P(b1 > 0) = 0.995$
F0	Robust Student-t	Stress, b1	2.30	1.14	[0.47, 4.16]	$P(b1 > 0) = 0.979$
F0	Gaussian	Recovery, b2	0.14	1.24	[-1.89, 2.11]	$P(b2 > 0) = 0.542$
F0	Robust Student-t	Recovery, b2	-0.39	1.14	[-2.21, 1.41]	$P(b2 > 0) = 0.368$

The residual scale parameter was lower in the robust model, median $\sigma = 6.52$, 89% CrI [5.93, 7.16]. Because this is the scale parameter of a Student-t likelihood rather than a Gaussian residual standard deviation, it should not be interpreted as directly equivalent to the Gaussian residual standard deviation.

NNE model. For NNE, the original Gaussian model estimated a negative stress effect, median = -0.65 dB, MAD = 0.28, 89% CrI [-1.10, -0.21], with $P(b1 < 0) = 0.990$. The robust Student-t model produced a somewhat larger negative estimate, median = -0.79 dB, MAD = 0.27, 89% CrI [-1.21, -0.36], with $P(b1 < 0) = 0.998$.

Thus, the NNE stress effect was not weakened by the robust specification. Instead, the robust model slightly strengthened the estimated negative stress-related change in NNE.

For the recovery contrast, the original Gaussian model estimated a small negative effect, median = -0.22 dB, MAD = 0.28, 89% CI [-0.67, 0.23], with $P(b2 > 0) = 0.219$. The robust model estimated a somewhat more negative effect, median = -0.32 dB, MAD = 0.25, 89% CI [-0.73, 0.09], with P(b2 > 0) = 0.105. Although directional evidence remained modest (pd = 0.895), the robust model shifted the recovery estimate in a more negative direction.

Outcome	Model	Parameter	Median	MAD	89% CI	Directional probability
NNE	Gaussian	Stress, b1	-0.65	0.28	[-1.10, -0.21]	P(b1 < 0) = 0.990
NNE	Robust Student-t	Stress, b1	-0.79	0.27	[-1.21, -0.36]	P(b1 < 0) = 0.998
NNE	Gaussian	Recovery, b2	-0.22	0.28	[-0.67, 0.23]	P(b2 > 0) = 0.219
NNE	Robust Student-t	Recovery, b2	-0.32	0.25	[-0.73, 0.09]	P(b2 > 0) = 0.105

The Student-t residual scale for NNE was median = 1.44, 89% CrI [1.28, 1.60]. As for F0, this parameter indexes the scale of the Student-t residual distribution and is not directly comparable to the Gaussian residual standard deviation.

Predictive comparison using PSIS-LOO. The robust models improved the approximate leave-one-observation-out cross-validation diagnostics. In the original Gaussian models, some observations had problematic Pareto-k values, indicating sensitivity to influential observations. In contrast, both robust Student-t models had all Pareto-k estimates below 0.7, indicating reliable PSIS-LOO estimates.

For F0, the robust model had better expected log predictive density than the Gaussian model. The LOO comparison favored the robust model by 16.3 elpd units, SE = 5.9. This suggests a clear improvement in expected out-of-sample predictive performance.

For NNE, the robust model was also preferred, with an elpd improvement of 12.9

units, SE = 6.9. This provides weaker but still meaningful evidence in favor of the robust specification.

Outcome	Preferred model	elpd difference for Gaussian vs robust	SE difference	Interpretation
F0	Robust Student-t	-16.3	5.9	Robust model clearly preferred
NNE	Robust Student-t	-12.9	6.9	Robust model moderately preferred

Here, negative elpd differences indicate worse expected predictive accuracy for the Gaussian model relative to the robust Student-t model.

Interpretation. Overall, the sensitivity analysis supports the robustness of the main inferential conclusions. The stress effect remained positive for F0 and negative for NNE under the robust Student-t specification. The F0 stress effect was attenuated but remained directionally strong and credibly positive. The NNE stress effect was slightly strengthened, with very high posterior probability for a negative effect.

The recovery effects were less stable and remained uncertain in both outcomes. For F0, the recovery effect was close to zero in the Gaussian model and slightly negative in the robust model, with directional probabilities near chance in both cases. For NNE, the recovery effect became more negative in the robust model, but uncertainty remained sufficient that this effect should be interpreted cautiously.

The robust models also improved predictive diagnostics. The elimination of problematic Pareto-k values suggests that the Student-t likelihood successfully reduced the influence of observations that were poorly handled by the Gaussian residual model. Because the substantive conclusions for the stress contrast were preserved while predictive diagnostics improved, the robust Student-t models provide a useful confirmation of the primary findings and may be preferred for sensitivity reporting.

Summary. As a sensitivity analysis, we refitted the main-effects models using a robust Student-t residual likelihood with $\nu = 4$ and correlated subject-level random effects. This specification was intended to assess whether the Gaussian models were sensitive to influential observations. For F0, the robust model estimated a positive stress effect, median = 2.30 Hz, MAD = 1.14, 89% CrI [0.47, 4.16], P(b1 > 0) = 0.979, compared with median = 3.27 Hz, MAD = 1.25, 89% CrI [1.25, 5.27], P(b1 > 0) = 0.995 in the Gaussian model. For NNE, the robust model estimated a negative stress effect, median = -0.79 dB, MAD = 0.27, 89% CrI [-1.21, -0.36], P(b1 < 0) = 0.998, compared with median = -0.65 dB, MAD = 0.28, 89% CrI [-1.10, -0.21], P(b1 < 0) = 0.990 in the Gaussian model. Recovery effects remained uncertain in both outcomes. PSIS-LOO comparisons favored the robust models over the Gaussian models for both F0, elpd difference = 16.3, SE = 5.9, and NNE, elpd difference = 12.9, SE = 6.9. Moreover, all Pareto-k estimates were below 0.7 in the robust models, indicating reliable LOO estimates. These results suggest that the main stress effects are robust to a heavy-tailed residual specification and that the Student-t models provide improved predictive diagnostics.

Personality Moderation of Vocal Stress Responses

Research Question. The central question addressed by this model is whether the effect of exam-related stress on vocal acoustics (F0, NNE) is moderated by individual differences in personality pathology. Specifically, we ask: Do the five PID-5 domains—Negative Affectivity, Detachment, Antagonism, Disinhibition, and Psychoticism—differentially amplify or attenuate the stress-induced changes in vocal parameters during (a) anticipatory stress and (b) post-stressor recovery?

The Challenge: Personality Traits from Intensive Longitudinal Data. A key methodological challenge in this study concerns how personality traits are represented in the moderation analysis. Each participant completed approximately 20 EMA assessments over 2.5 months, providing repeated measures of each PID-5 domain. A naive approach would aggregate these observations into a single person-level mean for each domain and use these means as predictors in a standard multilevel regression. However, this approach discards valuable information and fails to account for measurement error: the observed person-means are noisy estimates of the true latent traits, and treating them as known quantities underestimates uncertainty in the moderation effects.

Our model addresses this problem through a *joint measurement-and-outcome model* that simultaneously estimates latent personality traits from the EMA data and their moderating influence on vocal stress responses. This integrated approach has three key advantages:

1. **Measurement error correction**: Rather than using observed means as fixed predictors, the model estimates each participant's true latent trait score (θ_{id}) as a parameter, with appropriate uncertainty. This uncertainty propagates into the moderation estimates, yielding appropriately calibrated credible intervals.

2. **Borrowing strength across observations**: The repeated EMA measurements inform the latent trait estimates through a measurement model, allowing the model to distinguish stable trait variance from occasion-specific fluctuations.

3. **Coherent uncertainty quantification**: Because the latent traits and their effects are estimated jointly, the posterior distributions for moderation parameters (γ_1, γ_2) fully reflect uncertainty about both the traits themselves and their influence on vocal outcomes.

Model Specification. The model comprises two interconnected components: a *measurement model* for the EMA-based personality assessments and an *outcome model* for the vocal parameters.

Measurement Model (EMA). Let X_{nd} denote the observed score for participant i on domain d at EMA occasion n. The measurement model specifies:

$$X_{nd} \sim \text{Normal}(\theta_{i[n],d}, \sigma_d^{\text{ema}})$$

where θ_{id} is the latent true trait score for participant i on domain d, and σ_d^{ema} captures occasion-to-occasion variability (including both state fluctuations and measurement error). The latent traits are given standard normal priors:

$$\theta_{id} \sim \text{Normal}(0, 1)$$

This formulation treats each participant's approximately 20 EMA observations as repeated noisy indicators of a stable underlying trait, with the model learning both the trait estimates and the amount of occasion-level variability for each domain.

Outcome Model (Vocal Parameters). Let y_j denote the vocal outcome (F0 or NNE) for observation j, with participant index $i[j]$. Stress effects are parameterized using two orthogonal contrasts:

- c_1: Stress contrast (pre-exam vs. baseline);
- c_2: Recovery contrast (post-exam vs. pre-exam).

The outcome model specifies:

$$y_j \sim \text{Normal}(\mu_j, \sigma_y)$$

where the linear predictor μ_j includes fixed effects, random effects, and the crucial trait $\times$ contrast interactions:

$$\mu_j = \alpha + u_{i,1} + (\beta_1 + u_{i,2}) \cdot c_{1j} + (\beta_2 + u_{i,3}) \cdot c_{2j} + \sum_{d=1}^{5} \left[a_d \cdot \theta_{id} + \gamma_{1d} \cdot c_{1j} \cdot \theta_{id} + \gamma_{2d} \cdot c_{2j} \cdot \theta_{id}\right]$$

The parameters are:

- α: Grand mean (baseline vocal parameter);
- β_1, β_2: Population-average stress and recovery effects;
- $u_{i,1}$, $u_{i,2}$, $u_{i,3}$: Participant-specific random intercept and slopes;
- a_d: Main effect of trait d on baseline vocal level;
- γ_{1d}: **Stress moderation**—how trait d amplifies or attenuates the stress effect;
- γ_{2d}: **Recovery moderation**—how trait d shapes post-stressor trajectory.

The moderation parameters γ_{1d} and γ_{2d} are the quantities of primary theoretical interest. A positive γ_{1d} indicates that higher levels of trait d are associated with larger stress-induced changes in the vocal parameter.

Random Effects Structure. Participant-level random effects are specified using a non-centered parameterization for computational efficiency:

$$u_{i,k} = z_{i,k} \cdot \tau_k, \quad z_{i,k} \sim \text{Normal}(0, 1)$$

where τ_k are the random effect standard deviations. This structure allows for individual differences in baseline levels ($u_{i,1}$), stress reactivity ($u_{i,2}$), and recovery ($u_{i,3}$) beyond what is explained by the PID-5 traits.

Prior Specification. Priors were chosen to be weakly informative, incorporating domain knowledge about plausible parameter ranges while allowing the data to dominate inference (Table S10).

The priors on the moderation parameters (γ_{1d}, γ_{2d}) provide modest regularization, shrinking estimates toward zero in the absence of strong evidence. This helps guard against

Table S10

Prior specifications for the moderation model.

Parameter	Prior	Rationale
α	Normal(220, 30)	Centered on typical female F0 (Hz)
β_1, β_2	Normal(0, 10)	Allows stress effects up to ±20 Hz
a_d	Normal(0, 5)	Modest trait effects on baseline
γ_{1d}, γ_{2d}	Normal(0, 3)	Regularization toward zero
τ_k	Exponential(0.5)	Weakly informative for SDs
σ_y	Exponential(0.1)	Allows residual SD up to ∼10 Hz
σ_d^{ema}	Exponential(1)	Weakly informative for EMA variability

overfitting given the 10 moderation parameters (5 domains × 2 contrasts) being estimated.

Stan Implementation. The model was implemented in Stan. The complete code is available in the online repository accompanying this article.

Interpretation of Key Parameters. The model yields posterior distributions for 10 moderation parameters of primary interest:

- $\gamma_{1,\text{NegAff}}, \cdots, \gamma_{1,\text{Psych}}$: How each PID-5 domain moderates stress-induced vocal change (stress contrast × trait).
- $\gamma_{2,\text{NegAff}}, \cdots, \gamma_{2,\text{Psych}}$: How each domain moderates recovery-phase vocal change (recovery contrast × trait).

These parameters are expressed in the original units of the vocal outcome (Hz for F0, dB for NNE) per standard deviation of the latent trait. For example, $\gamma_{1,\text{NegAff}} = 3.0$ would indicate that a one-SD increase in latent Negative Affectivity is associated with an additional 3 Hz increase in F0 during the stress phase, beyond the population-average stress effect.

Posterior Predictive Checks. The `generated quantities` block produces posterior predictive samples (y_{rep}) for model checking. These were used to verify that the model adequately captured the distributional properties of the observed vocal data, including means, variances, and the pattern of individual differences across assessment phases.

Prior-Sensitivity Analysis

The probability of direction is, in principle, sensitive to the prior placed on the parameter under test, and the moderation magnitudes are only weakly constrained by the present sample. To make this dependence explicit, we refitted the moderation models under three prior standard deviations on the interaction coefficients (γ): half the default (1.5), the default (3.0), and double the default (6.0). We focus on the three effects with the strongest directional evidence in the primary analysis: Negative Affectivity × stress and Antagonism × recovery on F0, and Psychoticism × recovery on NNE.

Table S11

Prior-sensitivity analysis. Posterior median, 89% credible interval, and probability of direction (pd) for the three headline moderation effects under three prior standard deviations on the interaction coefficients (half, default, and double the default).

Effect	Prior SD	Default	Median	89% CrI	pd
Antagonism x recovery (F0)	1.5		1.62	[-0.21, 3.44]	.92
Antagonism x recovery (F0)	3.0	default	3.16	[0.62, 5.76]	.98
Antagonism x recovery (F0)	6.0		4.48	[1.39, 7.57]	.99
NegAff x stress (F0)	1.5		1.76	[-0.09, 3.62]	.93
NegAff x stress (F0)	3.0	default	3.14	[0.46, 5.85]	.97
NegAff x stress (F0)	6.0		4.39	[1.15, 7.64]	.98
Psychoticism x recovery (NNE)	1.5		0.80	[0.03, 1.55]	.95
Psychoticism x recovery (NNE)	3.0	default	0.88	[0.06, 1.68]	.96
Psychoticism x recovery (NNE)	6.0		0.90	[0.08, 1.73]	.96

Two regularities emerge. First, the *direction* of all three effects is robust: the probability of direction remains high under every prior, with an overall minimum of about .92 (Antagonism × recovery under the tightest prior). Second, the *magnitudes* are prior-dependent, and markedly so for the F0 effects. The Negative Affectivity × stress median moves from roughly 1.8 Hz under the tightest prior to about 4.4 Hz under the widest — an approximately 2.5-fold range — and its 89% interval includes zero under the tightest prior even though its *pd* stays near .94. The NNE effect (Psychoticism × recovery) is considerably more stable in magnitude (median ≈ 0.80–0.90 dB across priors, *pd* ≈ .95). All models converged cleanly ($\hat{R} \approx 1.00$, no divergent transitions). This is the empirical basis for the reporting strategy adopted throughout: we foreground direction and the probability of direction, and treat the reported magnitudes — and the widths of their credible intervals — as estimates carrying substantial, partly prior-dependent uncertainty.

Within-Person Temporal Covariation

We examined whether momentary fluctuations in personality states—assessed via EMA immediately before each voice recording—covaried with concurrent F0 levels. This tests whether trait-level moderation effects have within-person analogs: do individuals show higher F0 when they report elevated negative affectivity in the moment?

We compared hierarchical specifications with fixed effects (population-average slopes only) versus random slopes (allowing individual differences in within-person associations). The random slopes model had higher predictive accuracy than the fixed-effects specification ($R^2 = 35\%$ vs 2.5%; ELPD difference ≈ 40 points). By itself, however, this improvement in fit does not establish identifiable individual-level effects. With only three observations per person, the 89% credible intervals for individual slopes were very wide and almost always included zero, so the sign and magnitude of any single participant's slope are not identifiable from these data; we therefore do not interpret individual slopes one by one.

This pattern reflects a limitation of the sparse design rather than a substantive null. The between-subject standard deviations of the slopes (σ_β) are non-negligible for several domains, suggesting some heterogeneity, but they are themselves only weakly identified, so we do not attempt to characterize which individuals have larger or smaller slopes. Denser temporal sampling would be required to resolve individual-level within-person dynamics.

MCMC Convergence Diagnostics

Estimation Procedure

All Bayesian models were estimated using Hamiltonian Monte Carlo (HMC) via Stan (Carpenter et al., 2017), accessed through the cmdstanr package in R. For the primary F0 moderation model, we ran 4 independent chains with 2,000 warmup iterations and 6,000 sampling iterations per chain, yielding 24,000 total post-warmup draws. Sampler adaptation parameters were set conservatively to ensure reliable exploration of the posterior: `adapt_delta = 0.99` and `max_treedepth = 15`. The same estimation procedure was applied to the NNE moderation model.

Convergence Assessment

Convergence was assessed using standard diagnostics recommended for HMC estimation (Vehtari et al., 2021):

1. **Divergent transitions**: No divergent transitions were observed in either model, indicating that the sampler explored the posterior without encountering regions of high curvature that would compromise inference.

2. **Maximum treedepth**: No iterations exceeded the maximum treedepth, suggesting that the sampler did not require truncation of its trajectory length.

3. **Energy Bayesian Fraction of Missing Information (E-BFMI)**: All chains showed E-BFMI values well above the recommended threshold of 0.2, indicating adequate exploration of the energy distribution.

4. $\hat{R}$ **(R-hat)**: The potential scale reduction factor was computed for all parameters. For the F0 model, all $\hat{R}$ values for key parameters (fixed effects, moderation coefficients, variance components) were below 1.01, and no parameters across the full model exceeded 1.05. Table S11 reports $\hat{R}$ values for key parameters.

5. **Effective Sample Size (ESS)**: Both bulk-ESS and tail-ESS were computed for all parameters. Bulk-ESS, which reflects the efficiency of sampling for posterior means and medians, exceeded 4,000 for all key parameters. Tail-ESS, which reflects sampling efficiency for posterior quantiles, exceeded 3,000 for all key parameters. These values substantially exceed the minimum recommended threshold of 400 (Vehtari et al., 2021).

Visual Diagnostics

Trace plots for all key parameters showed excellent mixing across chains, with no visible trends or differences between chains. Representative trace plots are shown in Figure S1.

Autocorrelation plots indicated low autocorrelation at lag 1 and negligible autocorrelation beyond lag 10 for all parameters, consistent with the high effective sample sizes reported above.

Posterior Predictive Checks

Model adequacy was assessed using posterior predictive checks (Gelman et al., 2013). Figure S2 shows the posterior predictive density overlaid on the observed F0 distribution. The model accurately captured the location, spread, and shape of the observed data.

Additional posterior predictive checks confirmed that the model adequately recovered:

- The mean F0 across all observations (observed: 192.5 Hz; 95% PPC interval: [191.2, 193.8])
- The standard deviation of F0 (observed: 22.1 Hz; 95% PPC interval: [20.8, 23.4])
- The pattern of means across assessment periods (Baseline < Post-exam < Pre-exam)

Table S12

Posterior summary and convergence diagnostics for key parameters (F0 moderation model).

Parameter	Mean	SD	$\hat{R}$	ESS (bulk)	ESS (tail)
Intercept (α)	192.95	1.78	1.001	4490	8441
Stress effect (β_1)	3.43	1.33	1.000	38288	19497
Recovery effect (β_2)	-0.49	1.34	1.000	38539	20141
γ_1 Negative Affectivity	3.14	1.68	1.000	39739	20045
γ_1 Detachment	-0.36	1.68	1.000	37929	20076
γ_1 Antagonism	-0.10	1.61	1.000	38621	18464
γ_1 Disinhibition	-0.21	1.84	1.000	37799	19097
γ_1 Psychoticism	-0.26	1.71	1.000	38639	19968
γ_2 Negative Affectivity	-0.45	1.70	1.000	37632	19706
γ_2 Detachment	-2.02	1.71	1.000	37665	19891
γ_2 Antagonism	3.16	1.61	1.000	37825	20302
γ_2 Disinhibition	-0.18	1.87	1.000	34125	19595
γ_2 Psychoticism	-1.42	1.70	1.000	39361	20089
Residual SD (σ_y)	8.85	0.44	1.000	14363	17189
Random intercept SD (τ_1)	18.81	1.27	1.000	5855	11332
Random stress slope SD (τ_2)	1.33	1.14	1.000	12845	13041
Random recovery slope SD (τ_3)	1.65	1.43	1.000	10546	13300

Summary

All convergence diagnostics indicated that the MCMC sampler successfully explored the posterior distribution. The absence of divergent transitions, combined with $\hat{R}$ values below 1.01 and effective sample sizes exceeding 4,000, provides strong evidence that the posterior summaries reported in the main text are reliable. Posterior predictive checks confirmed that the model adequately captured the key features of the observed data.

NNE Model Diagnostics

The same estimation and diagnostic procedures were applied to the NNE moderation model. Table S12 summarizes the convergence diagnostics for both models.

Table S13

Convergence diagnostics summary for F0 and NNE moderation models.

Model	Divergences	Max Treedepth Exceeded	$\hat{R}$ (max)	$\hat{R} > 1.01$	ESS Bulk (min)	ESS Tail (min)
F0	0	0	1.0009	0	4413	8211
NNE	0	0	1.0039	0	3578	4466

The NNE model showed equally good convergence properties: no divergent transitions, no iterations exceeding maximum treedepth, all $\hat{R}$ values below 1.01, and minimum effective sample sizes well above recommended thresholds. Posterior predictive checks for the NNE model (Figure S3) confirmed adequate model fit.

Posterior Predictive Assurance of Moderation Effects

Rationale

Beyond estimating posterior distributions for the moderation parameters, we conducted a posterior predictive assurance analysis to quantify the probability that the *direction* of the observed moderation effect would replicate under the same study design. Rather than addressing statistical significance or interval exclusion criteria, this analysis focuses on directional replicability: given the fitted model and the uncertainty in its parameters, how likely is it that a new study with the same design would yield a moderation effect in the same direction?

This approach is conceptually distinct from frequentist power analysis. Instead of assuming a fixed but unknown true effect size, posterior predictive assurance integrates over the full posterior distribution of the model parameters, thereby reflecting uncertainty about both the magnitude of the effect and the data-generating process.

Procedure

Posterior predictive simulations were conducted using draws from the joint posterior distribution of the fitted moderation model. For each simulation, a new dataset was generated under the same design as the original study, including the same number of participants, the same stress and recovery contrasts, and a comparable distribution of EMA observations per participant. Latent personality traits for the new participants were drawn from their population prior distributions, consistent with the generative assumptions of the model.

For each simulated dataset, the moderation effect of interest—specifically, the interaction between Negative Affectivity and the stress contrast—was re-estimated using a fast proxy model. Replication success was defined using a minimal and directionally focused criterion: the estimated moderation effect was required to be positive (> 0). This criterion captures whether the effect would replicate in direction, without imposing additional thresholds related to statistical significance or effect size magnitude. This process was repeated 1,000 times, yielding an empirical estimate of the posterior predictive probability of directional replication (assurance).

Results

Across posterior predictive replications, the moderation effect was positive in 92.1% of simulated datasets. The estimated posterior predictive probability of replication success was therefore 0.92, with a 95% credible interval of [0.90, 0.94], reflecting Monte Carlo uncertainty in the simulation-based estimate.

Interpretation

These results indicate a high probability that the direction of the moderation effect would replicate in a new sample drawn under the same design assumptions. In other words, given the fitted model and the uncertainty in its parameters, the interaction between Negative Affectivity and stress is expected to be positive in the large majority of replications.

At the same time, this analysis does not imply precise recovery of the effect magnitude. The posterior distribution of the moderation parameter remains relatively wide, indicating substantial uncertainty about the exact size of the effect. The assurance analysis therefore supports *directional robustness* of the moderation effect, while remaining agnostic about the degree of precision with which its magnitude can be estimated.

Taken together with the posterior estimates reported in the main analysis, these results suggest that the observed moderation effect is unlikely to be a chance reversal in direction, even though its quantitative strength should be interpreted with appropriate caution.

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